

CNN-Mamba-k7

From a two-head baseline to frame-aware auxiliary learning

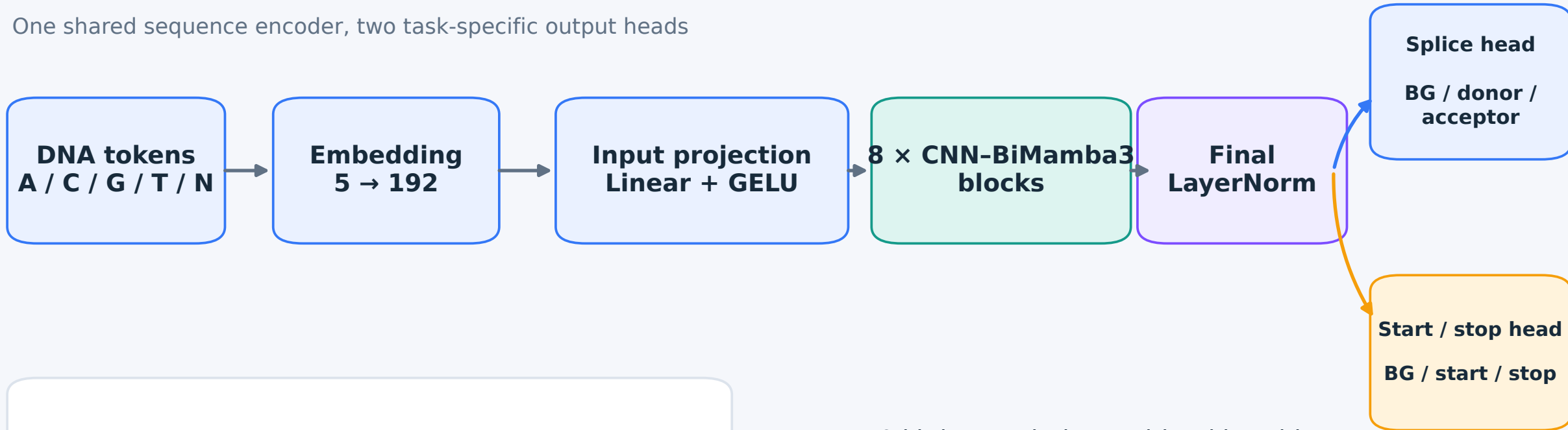
Drosophila • 10 kb windows • splice sites and translation boundaries



Architecture, supervision, inference, and measured impact

Previous model: CNN-Mamba-k7 baseline

One shared sequence encoder, two task-specific output heads



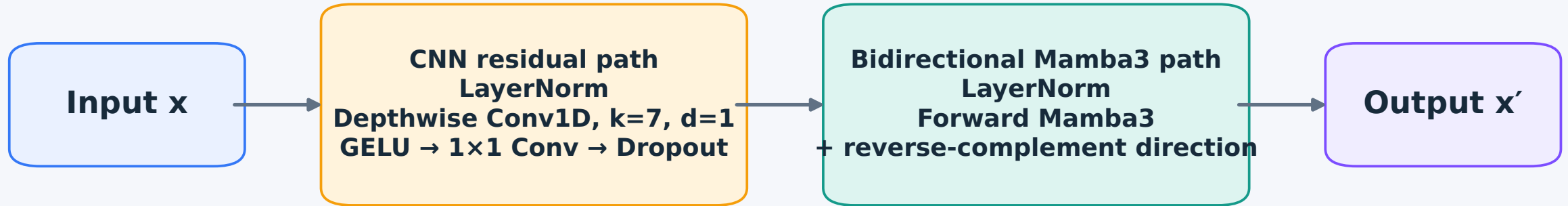
Training objective

$$L = 0.25 \times L_{\text{splice}} + 1.0 \times L_{\text{start/stop}}$$

- 10 kb input windows with 5 kb stride
- Canonical-candidate cross-entropy
- No explicit reading-frame target
- Start and stop must be learned indirectly from sequence context

Inside each baseline CNN-BiMamba3 block

Local motif extraction and long-range bidirectional context



Residual updates preserve the original representation at both stages.

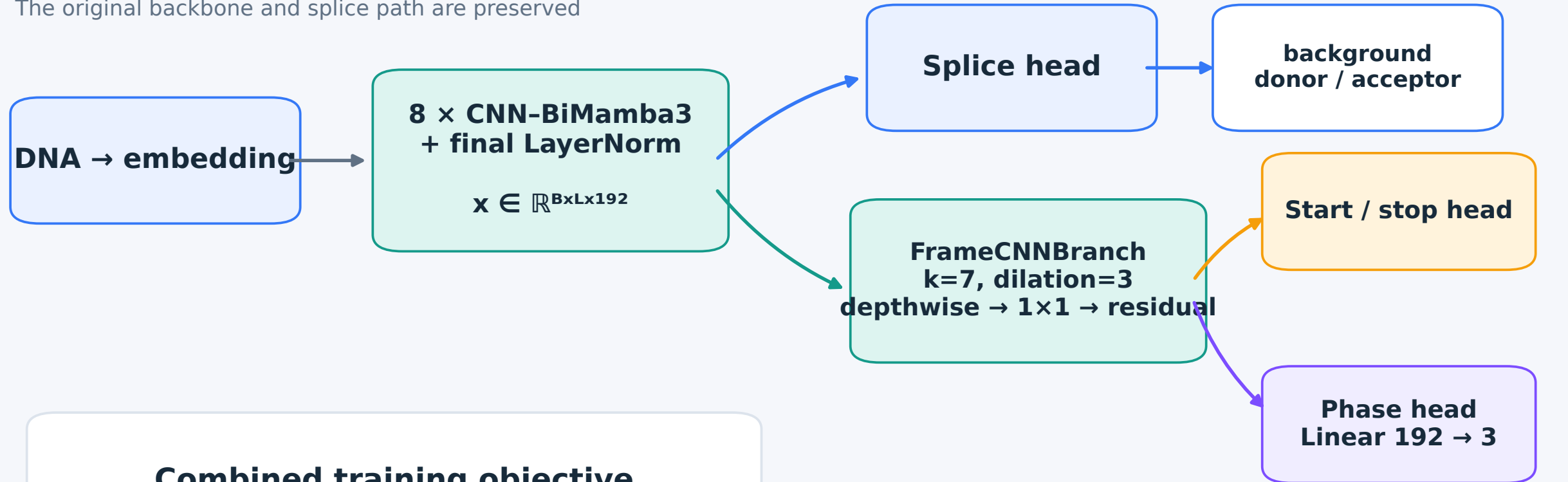
k=7, dilation=1 samples adjacent positions



- Strong splice-site performance from local motifs + long context
- Adjacent CNN taps mix all three codon offsets
- No auxiliary pressure to represent coding phase

Current model: explicit frame-aware auxiliary learning

The original backbone and splice path are preserved



Combined training objective

$$L = 0.25L_{\text{splice}} + L_{\text{start/stop}} + \lambda_{\text{phase}} L_{\text{phase}}$$

- Start/stop and phase share frame-aware features
- Splice predictions bypass the added frame branch
- Phase head is auxiliary: used for training, not required by UniAnn

What dilation = 3 changes

Same output length; different sampling pattern

Baseline $k=7$, dilation=1

effective receptive field: 7 bp



Frame branch $k=7$, dilation=3

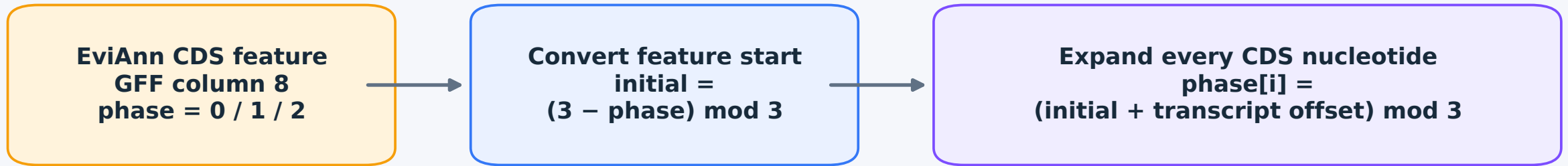
effective receptive field: 19 bp



Taps are spaced by 3 bp: $i-9$, $i-6$, $i-3$, i , $i+3$, $i+6$, $i+9$.
They preserve the same codon offset without striding or downsampling.

From local GFF phase to per-base supervision

The annotation is expanded; it is never supplied as model input



Example

A	T	G	G	A	A	T	A	A
0	1	2	0	1	2	0	1	2

per-base phase target

- Reverse strand is converted in transcript direction
- Outside CDS: masked
- Conflicting isoform phases: masked
- All 115,688 EviAnn CDS junctions passed frame-continuity checks

Each 10 kb sample stores `phase_labels[10,000]` and `phase_mask[10,000]`.
Only mask = 1 positions contribute to cross-entropy.

Training and inference are intentionally different

Phase annotation shapes the representation but is not required at prediction time

TRAINING

DNA sequence → three outputs

- splice logits
- start/stop logits
- phase logits

Phase labels contribute $\lambda_{\text{phase}} \times \text{CE}$

trained weights



INFERENCE

DNA sequence → two outputs

- splice probabilities
- start/stop probabilities

No phase labels; phase logits are not requested

Gradient path

**L_{phase} → phase head → frame branch
→ shared CNN-Mamba backbone**

Downstream compatibility

**Same six-column score file
Same UniAnn interface and thresholds**

The model learns frame-aware features; it does not receive the true frame at inference.

Held-out chrX ablation: where the gain comes from

Exact AUPRC over every canonical candidate on the complete chrX positive strand

Model	Donor	Acceptor	Start	Stop	Mean
Baseline	.9472	.9454	.7176	.5337	.7860
Dilation-3 only	.9438	.9417	.7549	.5888	.8073
Phase auxiliary	.9516	.9468	.8412	.8760	.9039
Combined	.9559	.9527	.8510	.8823	.9105

Phase supervision is the main driver

Stop: 0.5337 → 0.8760

Dilation-3 adds a smaller but consistent frame-aware gain

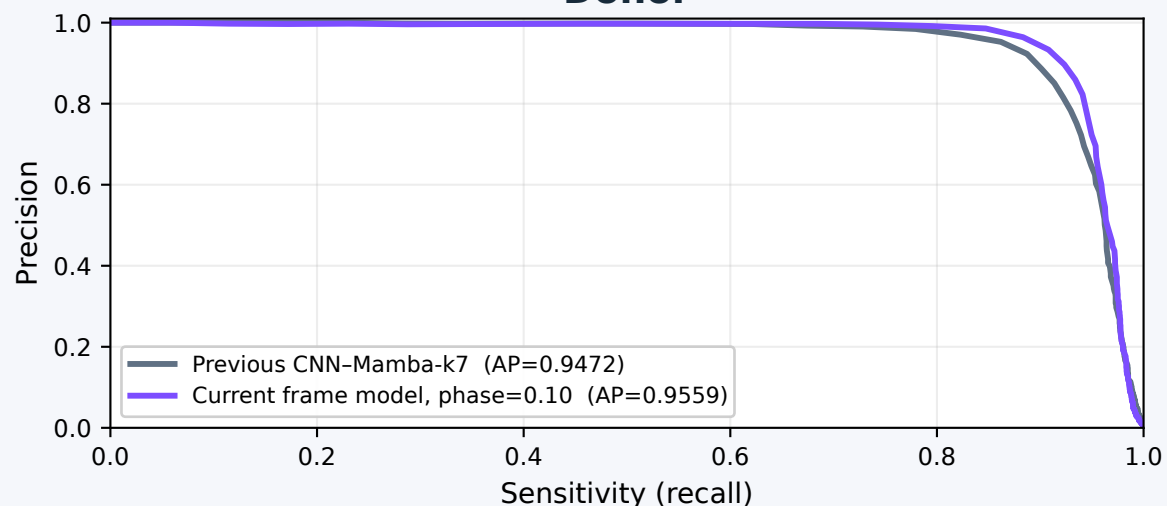
Combined mean: 0.9105

No reference annotation from chrX was used for training.

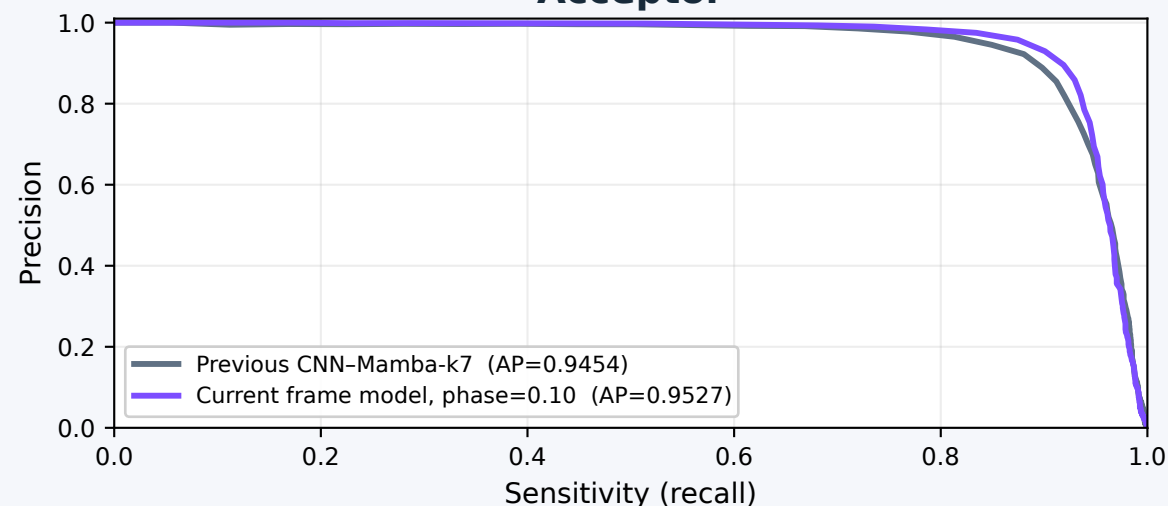
Four-task PR-sensitivity curves: held-out chrX

Honest generalization: chrX excluded from model training

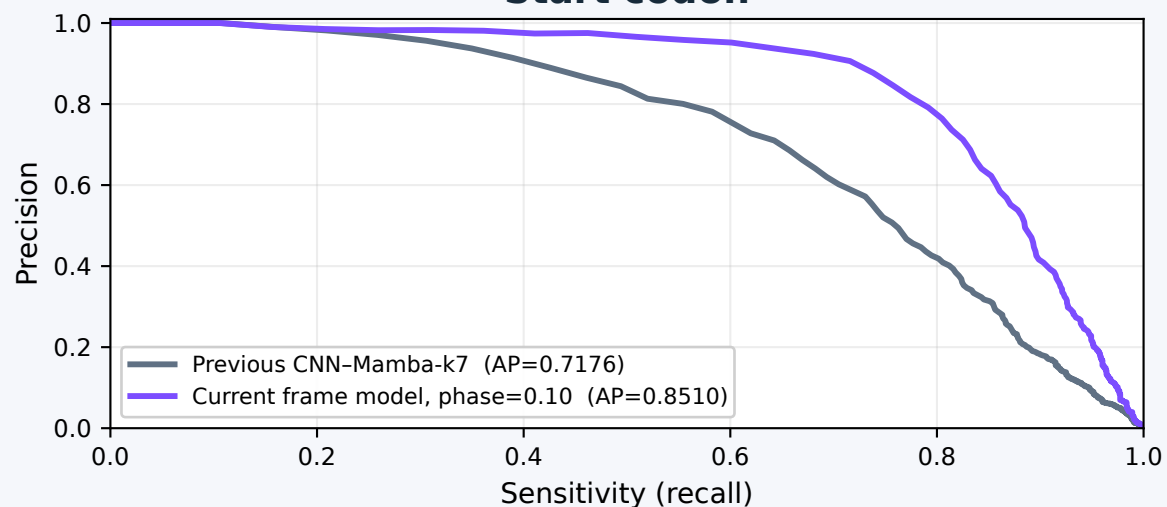
Donor



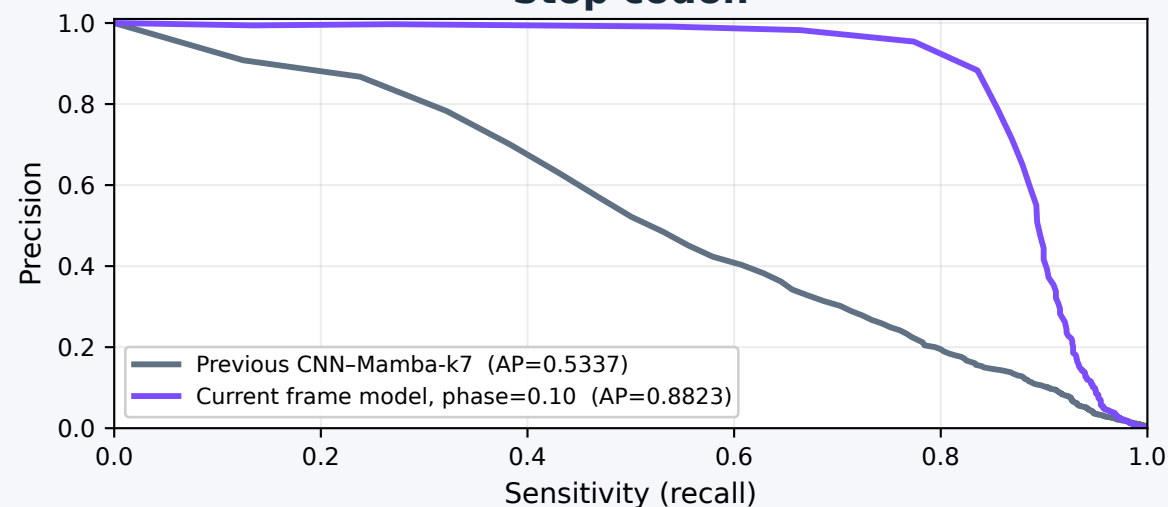
Acceptor



Start codon



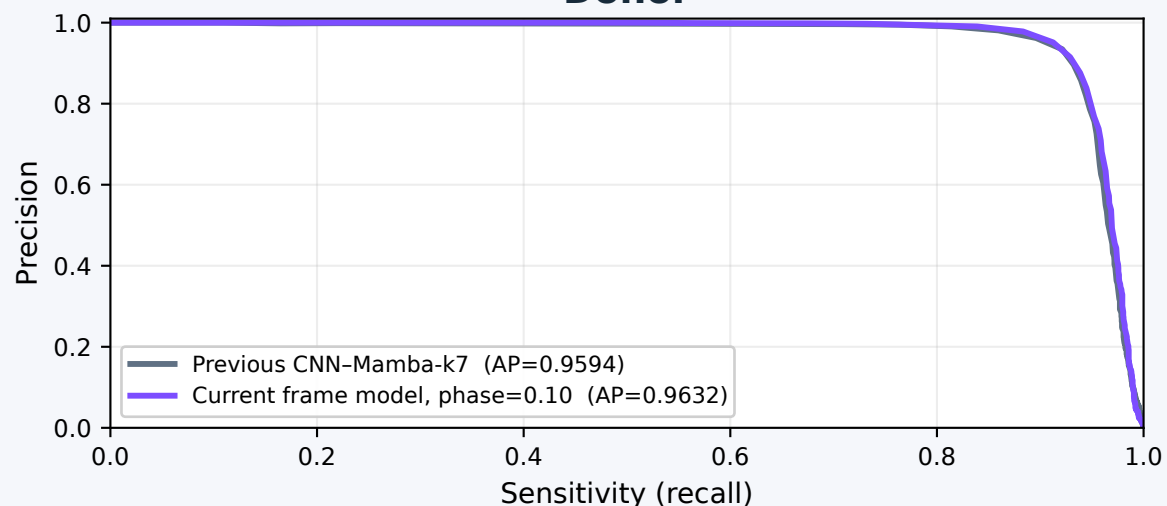
Stop codon



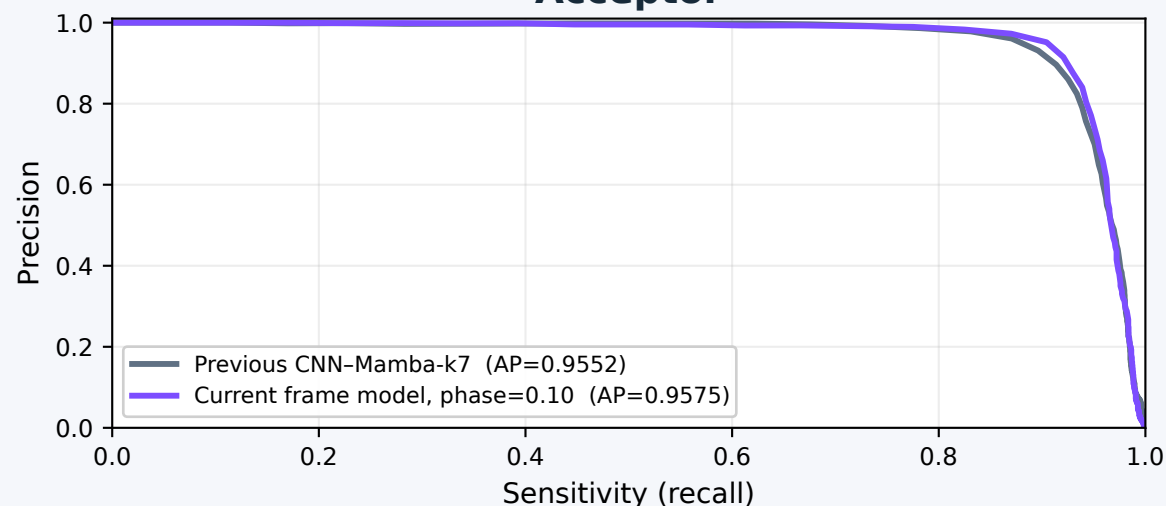
Four-task PR-sensitivity curves: chrX-in-training

Diagnostic fit: chrX candidates come from a chromosome included in training

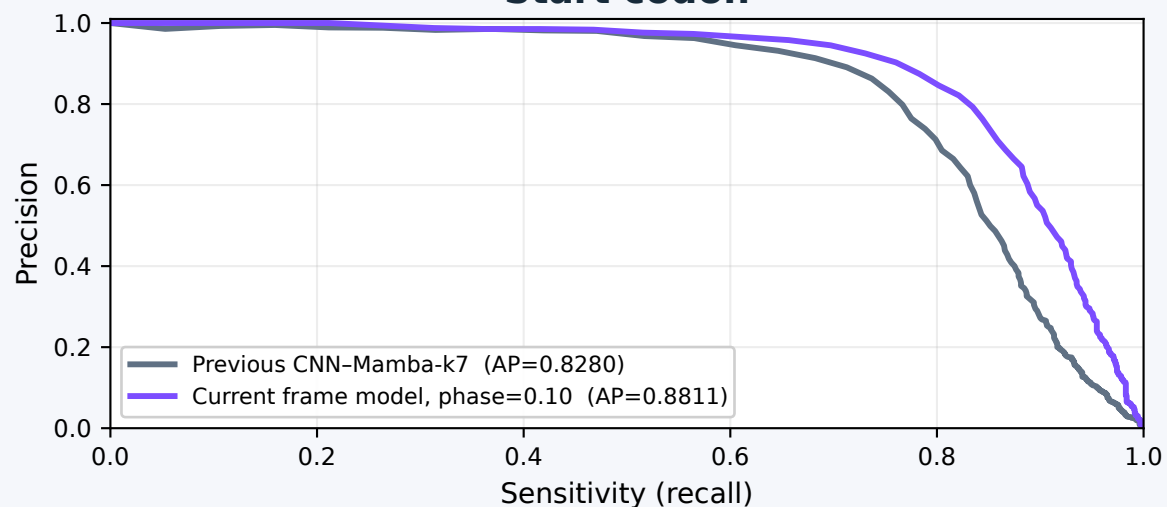
Donor



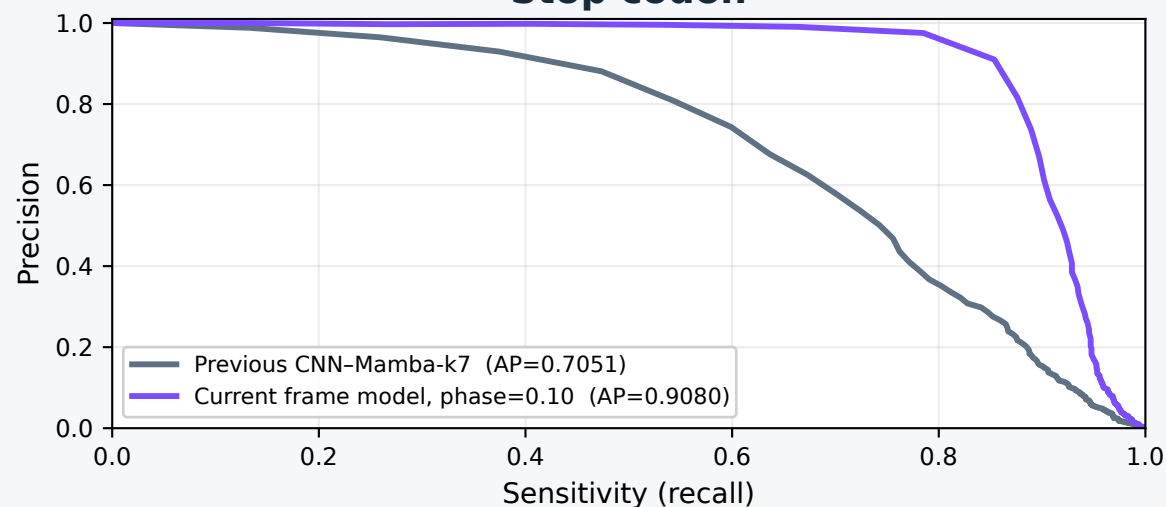
Acceptor



Start codon



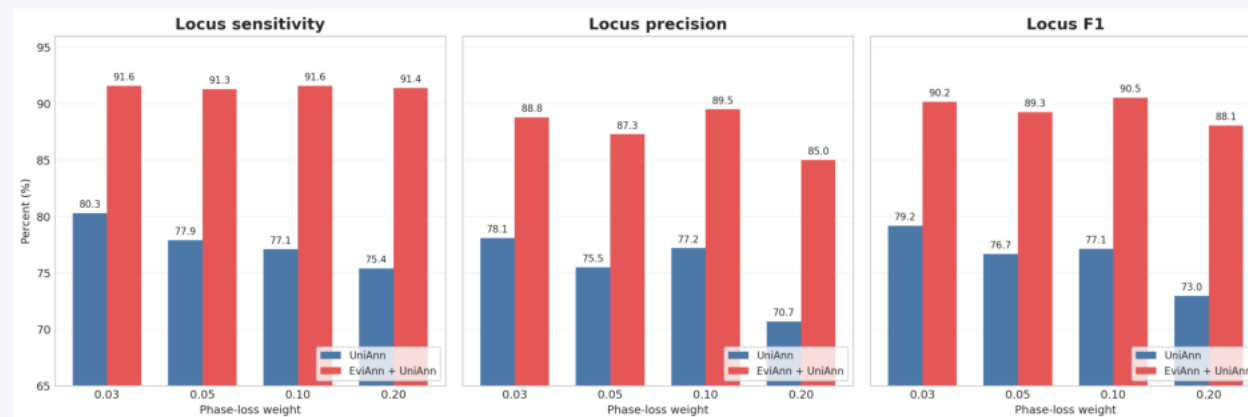
Stop codon



Phase-loss coefficient: no single value wins every metric

dilation=3 fixed; only λ_{phase} changes

λ_{phase}	Held-out mean AP	In-train mean AP	UniAnn F1	+ EviAnn F1
0.03	.9056	.9334	79.18	90.18
0.05	.9105	.9272	76.68	89.26
0.10	.9105	.9274	77.15	90.54
0.20	.9024	.9100	72.97	88.08

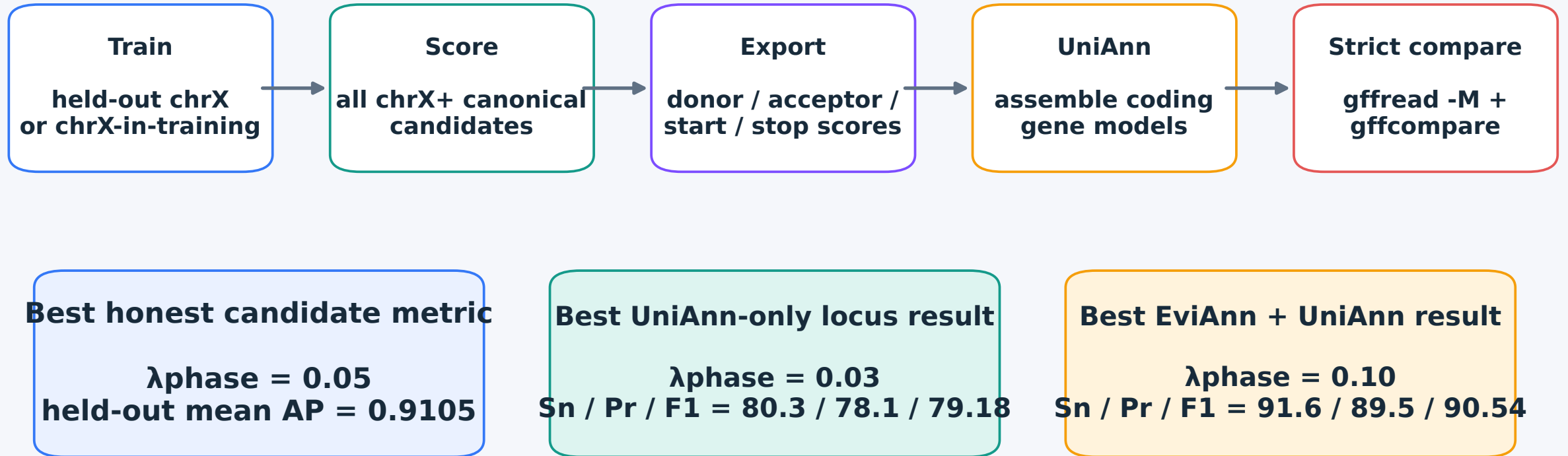


Pareto choices
0.03: UniAnn-only
0.05: held-out AP | 0.10: combined locus F1

0.20 is consistently worse.
0.05 and 0.10 held-out mean AP are effectively tied.

End-to-end evaluation and practical takeaway

Model quality is evaluated at both candidate and gene-model levels



Recommended name: CNN-Mamba-k7 with a dilation-3 frame branch and phase auxiliary supervision

The architecture is fixed across the coefficient grid; only the phase-loss weight changes.